Deva Anand

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Education

University of Massachusetts Amherst — **September 2025 – May 2027 (Expected)**
Master of Science in Computer Science — Amherst, MA
Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies — **August 2019 – May 2023**
Bachelor of Engineering in Computer Science and Engineering — Chennai, India

Publication

- *"Alzheimer's Disease Classification using Transfer Learning,"* **IEEE CONIT 2023** (3rd International Conference on Intelligent Technologies), applied deep transfer learning on neuroimaging data for medical image classification (first author).

Skills

Languages	Python, C++, JavaScript, TypeScript, SQL, Bash
Generative AI / Deep Learning	PyTorch, JAX, HuggingFace Transformers, NumPy, normalizing flows, diffusion (DDPM), variational inference, transfer learning, from-scratch autodiff, mechanistic interpretability
LLMs / RAG / Retrieval	LLM extraction & evaluation pipelines, RAG, RRF, cross-encoder reranking, MMR, vector embeddings (sentence-transformers, FAISS), llama-index, spaCy, Ollama, OpenAI / Groq APIs
Data & Pipelines	Apache Spark / PySpark, Spark UI, large-scale ETL, FastAPI, Node.js, REST APIs, MongoDB, PostgreSQL, Redis, SQLite
Cloud & DevOps	AWS, Azure, GCP, Docker, Git, GitHub Actions, CI/CD

Research & Projects

Generative Models & Deep Learning — github.com/Deva-1903/UMASS_CICS_689
JAX, PyTorch, NumPy | UMass CS 689 Advanced Machine Learning
- Implemented **generative AI models from scratch** in JAX, a **normalizing flow with coupling layers** and a **DDPM diffusion model**, deriving the ELBO and closed-form Gaussian-KL training objectives via variational inference.
- Built a matrix-based **reverse-mode autodiff engine from scratch in NumPy** (validated against JAX to machine precision) and benchmarked **5 neural architectures** (perceptron, MLPs, VGG-style CNN, ResNet) across **3 optimizers** on CIFAR-10.

Refusal Decay - LLM Safety & Mechanistic Interpretability — github.com/Deva-1903/refusal-decay
PyTorch, HuggingFace, Llama-3.1-8B | UMass COMPSCI 602 Research (graded full marks)
- Studied how prefilling attacks weaken refusal behavior in a safety-aligned LLM, extracting the residual-stream **refusal direction** via difference-in-means and tracing it per layer as behavioral refusal dropped **0.92 to 0.32**, evidence relevant to AI safety and governance.
- Designed causal interventions (activation patching, direction injection via forward hooks) with random/orthogonal controls and reported a clean null (**0/300** recovered) with bootstrap 95% CIs and McNemar's test.

KG2RAG-Enhanced - Multi-Hop QA on HotpotQA — github.com/Deva-1903/KG2RAG-685-NLP
Python, Ollama + LLaMA-3, sentence-transformers, llama-index, spaCy | UMass CS 685 NLP
- Led the **multi-view retrieval** component of a team KG-guided **RAG** pipeline, query-side sub-question decomposition, per-view dense retrieval, RRF fusion, cross-encoder reranking, and MMR, lifting supporting-fact recall **+2.68%** (54.53 to 57.21) over the KG^{2}RAG baseline on a **4,905-question** evaluation.

AgenticSearch - Provenance-First Entity Discovery — github.com/Deva-1903/ciir_agentic_search
Python, FastAPI, OpenAI / Groq, Brave Search, SQLite
- Built a multi-stage **LLM** discovery pipeline that extracts structured entities with deterministic parsers before LLM fallback and returns **cell-level provenance** per value; hardened with **150+ automated tests** and provider routing across backends.

Experience

Wysa — **July 2023 – July 2025**
Backend Engineer — Bengaluru, India
- Architected a full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label data for production clinical NLP models; replaced spreadsheet workflows, cutting manual annotation **100+ hours/month** and improving labeling throughput **60%**.
- Engineered a **multi-tenant NHS eTriage** healthcare backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability over sensitive clinical data and remediated **20+ VAPT findings**.
- Participated in a measured **GitHub Copilot** productivity pilot, routing tech-debt PRs, code migrations, and security-finding triage through AI-assisted workflows and maintaining a versioned conventions file to keep AI-generated code aligned with team patterns.
- Implemented **AWS KMS** envelope encryption for clinical PII and enforced a **90-day data-retention** policy per NHS / client agreement, applying healthcare data-governance and compliance practices.

Extracurriculars

- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.